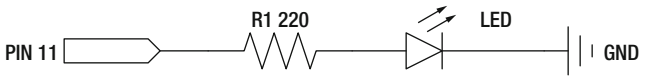

Figure 5-2. *An example of sourcing current*

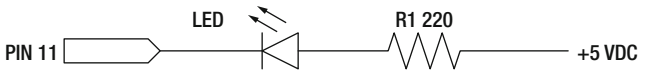

Figure 5-3. *An example of sinking current*

In comparing these two schematics, you should notice there is a difference in whether or not one side of the drawing is connected to +5 VDC or to GND and that this connection affects the direction of the LED. LEDs and a few other components are **polarized**, meaning that they only work in one direction, or with the correct polarity. By sourcing current, the Arduino I/O pin acts as source for +5 volts, providing a high-side positive supply of current. On the other hand, by sinking current, the Arduino I/O pin acts like a connection to ground, providing a low-side or negative current.

While each pin can provide about 40 milliamps of current, this is just about enough juice for one or two LEDs to work on each pin safely. Drawing more than 40 milliamps or exceeding +5v on any pin may either damage the microcontroller or the connected device. To safeguard against this happening, it is generally a good idea to connect output pins to other devices using one of two components, the first of which is a **current limiting resistor** used to drop or limit the current needed by the device. Playing it safe, a 1 kilohm resistor is a common value for current limiting for general applications, although we can get away with less resistance, as with our earlier LED schematics where we used a 220 ohm resistor to get more brightness out of the LED without damaging anything. Alternatively, we could use **transistors** or other switching components to switch larger current loads, like motors, solenoids, or strings of LEDs that exceed the maximum 40 milliamps of current and/or operate at voltages greater than +5v.

■ **Note** While it is possible to damage the microcontroller from connecting things wrong, it is a surprisingly tough little chip and is fairly hard to completely ruin. More often than not, it will be okay after removing power to it and disconnecting the offending circuit. Sometimes, however, it is possible to end up with a dead pin that will no longer work, but only in catastrophic situations will the entire chip release the magic smoke.